Week 1: HTML structure

The page skeleton

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport"
        content="width=device-width, initial-scale=1">
  <title>Page title (shows in the tab)</title>
</head>
<body>
  ...everything the visitor sees...
</body>
</html>
```

head = information about the page. **body** = the page itself. Tags come in pairs: `<p>` opens, `</p>` closes. Some tags have no closing tag: `<br>`, `<hr>`, `<img>`, `<meta>`.

Text

Tag	Use
`<h1>` to `<h6>`	Headings. One `h1` per page; do not skip levels
`<p>`	Paragraph
`<strong>` / `<em>`	Important (bold) / emphasis (italic)
` ` / `<hr>`	Line break / horizontal line
`<ul>` `<ol>` `<li>`	Bulleted list / numbered list / list item
`<div>` / `<span>`	Generic block / inline container (use when no better tag exists)

Links and images

```
<a href="about.html">About us</a>
<a href="https://example.com">Another site</a>
<img src="images/cat.png" alt="A grey cat asleep">
```

- **Every image needs `alt` text.** It is read aloud by screen readers and shown if the image fails to load.
- `href` = where a link goes. `src` = where an image comes from.

File paths (where is the file?)

Path	Means
`page.html`	Same folder as this file
`images/a.png`	Inside the sub-folder `images`
`../index.html`	One folder up, then the file
`https://...`	A whole address on the web

File names: lowercase, no spaces (`my-page.html`). The home page of a site is usually called `index.html`.

Semantic layout tags

```
<header>  site title, logo        </header>
<nav>     main links               </nav>
<main>    the unique content       </main>
<section> a themed part of a page  </section>
<article> content that stands alone</article>
<aside>   side information         </aside>
<footer>  author, contact          </footer>
```

They look the same as `<div>` on screen, but they **tell browsers, search engines and screen readers what each part is**.

Attributes and special characters

- `id` is unique on the page. `class` can be shared by many tags. `lang` sets the language.
- Write `<` for `<`, `>` for `>`, `&` for `&`, `"` for a quote, `©` for the copyright sign.
- Attribute values go in quotes: `class="card"`.

Preview in Codespaces

```
python3 -m http.server 8000
# then open the forwarded port 8000
```

Save the file, then refresh the preview. Stop the server with Ctrl+C.

Top mistakes

- A missing closing tag or a missing quote: the page looks wrong from that point.
- Wrong file path: broken image icon or a "404 Not Found".
- Opening a tag in one order and closing in another. Close the **inner** tag first: `<p><em>x</em></p>`.
- Using `<br>` many times to make space. Use CSS (Week 2) instead.
- Skipping `alt`. Always describe the picture.